

Implicit Differentiation

IMPLICIT FUNCTIONS. An equation $f(x, y) = 0$, on perhaps certain restricted ranges of the variables, is said to define y *implicitly* as a function of x .

EXAMPLE 1: (a) The equation $xy + x - 2y - 1 = 0$, with $x \neq 2$, defines the function $y = \frac{1-x}{x-2}$.

(b) The equation $4x^2 + 9y^2 - 36 = 0$ defines the function $y = \frac{2}{3}\sqrt{9-x^2}$ when $|x| \leq 3$ and $y \geq 0$, and the function $y = -\frac{2}{3}\sqrt{9-x^2}$ when $|x| \leq 3$ and $y \leq 0$. The ellipse determined by the given equation should be thought of as consisting of two arcs joined at the points $(-3, 0)$ and $(3, 0)$.

The derivative y' may be obtained by one of the following procedures:

1. Solve, when possible, for y and differentiate with respect to x . Except for very simple equations, this procedure is to be avoided.
2. Thinking of y as a function of x , differentiate both sides of the given equation with respect to x and solve the resulting relation for y' . This differentiation process is known as *implicit differentiation*.

EXAMPLE 2: (a) Find y' , given $xy + x - 2y - 1 = 0$.

We have $x \frac{d}{dx}(y) + y \frac{d}{dx}(x) + \frac{d}{dx}(x) - 2 \frac{d}{dx}(y) - \frac{d}{dx}(1) = \frac{d}{dx}(0)$
or $xy' + y + 1 - 2y' = 0$; then $y' = \frac{1+y}{2-x}$.

(b) Find y' when $x = \sqrt{5}$, given $4x^2 + 9y^2 - 36 = 0$.

We have $4 \frac{d}{dx}(x^2) + 9 \frac{d}{dx}(y^2) = 8x + 9 \frac{dy}{dy}(y^2) \frac{dy}{dx} = 8x + 18yy' = 0$
or $y' = -4x/9y$. When $x = \sqrt{5}$, $y = \pm 4/3$. At the point $(\sqrt{5}, 4/3)$ on the upper arc of the ellipse, $y' = -\sqrt{5}/3$, and at the point $(\sqrt{5}, -4/3)$ on the lower arc, $y' = \sqrt{5}/3$.

DERIVATIVES OF HIGHER ORDER may be obtained in two ways. The first is to differentiate implicitly the derivative of one lower order and replace y' by the relation previously found.

EXAMPLE 3: From Example 2(a), $y' = \frac{1+y}{2-x}$. Then

$$\frac{d}{dx}(y') = y'' = \frac{d}{dx}\left(\frac{1+y}{2-x}\right) = \frac{(2-x)y' + 1+y}{(2-x)^2} = \frac{(2-x)\left(\frac{1+y}{2-x}\right) + 1+y}{(2-x)^2} = \frac{2+2y}{(2-x)^2}$$

The second method is to differentiate implicitly both sides of the given equation as many times as is necessary to produce the required derivative and eliminate all derivatives of lower order. This procedure is recommended only when a derivative of higher order at a given point is required.

EXAMPLE 4: Find the value of y'' at the point $(-1, 1)$ of the curve $x^2y + 3y - 4 = 0$.

We differentiate implicitly with respect to x twice, obtaining

$$x^2y' + 2xy + 3y' = 0 \quad \text{and} \quad x^2y'' + 2xy' + 2xy' + 2y + 3y'' = 0$$

We substitute $x = -1$, $y = 1$ in the first relation to obtain $y' = \frac{1}{2}$. Then we substitute $x = -1$, $y = 1$, $y' = \frac{1}{2}$ in the second relation to get $y'' = 0$.

Solved Problems

1. Find y' , given $x^2y - xy^2 + x^2 + y^2 = 0$.

$$\begin{aligned}\frac{d}{dx}(x^2y) - \frac{d}{dx}(xy^2) + \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= 0 \\ x^2 \frac{d}{dx}(y) + y \frac{d}{dx}(x^2) - x \frac{d}{dx}(y^2) - y^2 \frac{d}{dx}(x) + \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= 0\end{aligned}$$

Hence $x^2y' + 2xy - 2xyy' - y^2 + 2x + 2yy' = 0$ and $y' = \frac{y^2 - 2x - 2xy}{x^2 + 2y - 2xy}$

2. Find y' and y'' , given $x^2 - xy + y^2 = 3$.

$$\frac{d}{dx}(x^2) - \frac{d}{dx}(xy) + \frac{d}{dx}(y^2) = 2x - xy' - y + 2yy' = 0. \quad \text{So} \quad y' = \frac{2x - y}{x - 2y}$$

$$\begin{aligned}\text{Then } y'' &= \frac{(x - 2y) \frac{d}{dx}(2x - y) - (2x - y) \frac{d}{dx}(x - 2y)}{(x - 2y)^2} = \frac{(x - 2y)(2 - y') - (2x - y)(1 - 2y')}{(x - 2y)^2} \\ &= \frac{3xy' - 3y}{(x - 2y)^2} = \frac{3x\left(\frac{2x - y}{x - 2y}\right) - 3y}{(x - 2y)^2} = \frac{6(x^2 - xy + y^2)}{(x - 2y)^3} = \frac{18}{(x - 2y)^3}\end{aligned}$$

3. Find y' and y'' , given $x^3y + xy^3 = 2$ and $x = 1$.

We have

$$x^3y' + 3x^2y + 3xy^2y' + y^3 = 0$$

and $x^3y'' + 3x^2y' + 3x^2y' + 6xy + 3xy^2y'' + 6xy(y')^2 + 3y^2y' + 3y^2y' = 0$

When $x = 1$, $y = 1$; substituting these values in the first derived relation yields $y' = -1$. Then substituting $x = 1$, $y = 1$, $y' = -1$ in the second relation yields $y'' = 0$.

Supplementary Problems

4. Establish formula 10 of Chapter 10 for $m = p/q$, p and q integers, by writing $y = x^{p/q}$ as $y^q = x^p$ and differentiating with respect to x .
5. Find y'' , given (a) $x + xy + y = 2$; (b) $x^3 - 3xy + y^3 = 1$.
- Ans. (a) $y'' = \frac{2(1+y)}{(1+x)^2}$; (b) $y'' = -\frac{4xy}{(y^2-x)^3}$
6. Find y' , y'' , and y''' at (a) the point $(2, 1)$ on $x^2 - y^2 - x = 1$; (b) the point $(1, 1)$ on $x^3 + 3x^2y - 6xy^2 + 2y^3 = 0$. Ans. (a) $3/2$, $-5/4$, $45/8$; (b) 1 , 0 , 0
7. Find the slope at the point (x_0, y_0) of (a) $b^2x^2 + a^2y^2 = a^2b^2$; (b) $b^2x^2 - a^2y^2 = a^2b^2$; (c) $x^3 + y^3 - 6x^2y = 0$.
- Ans. (a) $-\frac{b^2x_0}{a^2y_0}$; (b) $\frac{b^2x_0}{a^2y_0}$; (c) $\frac{4x_0y_0 - x_0^2}{y_0^2 - 2x_0^2}$
8. Prove that the lines tangent to the curves $5y - 2x + y^3 - x^2y = 0$ and $2y + 5x + x^4 - x^3y^2 = 0$ at the origin intersect at right angles.

9. (a) The total surface area of a rectangular parallelepiped of square base y on a side and height x is given by $S = 2y^2 + 4xy$. If S is constant, find dy/dx without solving for y .
(b) The total surface area of a right circular cylinder of radius r and height h is given by $S = 2\pi r^2 + 2\pi rh$. If S is constant, find dr/dh .

Ans. (a) $-\frac{y}{x+y}$; (b) $-\frac{r}{2r+h}$

10. For the circle $x^2 + y^2 = r^2$, show that $\left| \frac{y''}{[1 + (y')^2]^{3/2}} \right| = \frac{1}{r}$.

11. Given $S = \pi x(x + 2y)$ and $V = \pi x^2 y$, show that $dS/dx = 2\pi(x - y)$ when V is a constant and $dV/dx = -\pi x(x - y)$ when S is a constant.